

6.2.2 Ionising radiation and risk

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) the nature and effects of ionising radiations: differences in ionising and penetrating power, effects on living tissue	
(ii) the stability and decay of nuclei in terms of binding energy; transformation of nucleus on emission of radiation; qualitative variation of binding energy with proton and neutron number ("Nuclear Valley")	HSW2, 8
(iii) nuclear fission; chain reaction; nuclear fusion; nuclear power generation.	Benefits and concerns over nuclear power HSW12
(b) <i>Make appropriate use of:</i>	
(i) the terms: nucleon number, proton number, isotope, binding energy, atomic mass unit, absorbed and effective dose, risk	
<i>by sketching and interpreting:</i>	
(ii) plots of binding energy per nucleon of nuclei against nucleon number.	
(c) <i>Make calculations and estimates involving:</i>	
(i) activity of a sample of radioactive material (related to half-life or decay constant)	
(ii) absorbed dose in gray = energy deposited per unit mass	HSW9, 10
(iii) effective dose in sievert = absorbed dose in gray × quality factor	HSW9, 10
(iv) energy changes from nuclear transformations: $E_{\text{rest}} = mc^2$.	